Lebesgue Dominated Convergence Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of lebesgue dominated convergence theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RA-04 / Real Analysis / Report v1

I. TARGET AND IMPORT CONTRACT

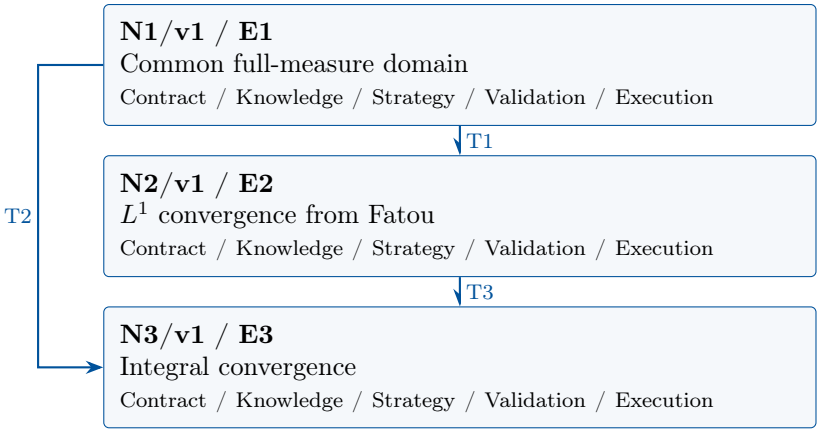
Theorem I.1. *On a measure space, let measurable $f_n \to f$ almost everywhere and $|f_n| \le g$ almost everywhere for all n, where $g \in L^1$ is nonnegative. Prove $f \in L^1$, $\|f_n - f\|_1 \to 0$, and $\int f_n \to \int f$. We use a measurable representative f of the almost-everywhere limit.*

A. Foundations and permitted imports

On a measure space (X, Σ, μ), Fatou's lemma says $\int \liminf h_n \le \liminf \int h_n$ for nonnegative measurable h_n. Use countable unions of null sets and the integral triangle inequality. For complex-valued functions, apply measurability to real and imaginary parts; the same absolute-value proof works.

Take f to be a measurable representative of the almost-everywhere limit; all exceptional sets are contained in a measurable null set. This convention avoids silently assuming the measure space is complete.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The limit and all terms are integrable; all bounds hold on one full-measure set.

Receiving activity. N2: L^1 convergence from Fatou

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Integrable f and bounds; $h_n = 2g - |f_n - f|$; prior exports E1/v1

T2: N1/E1/v1 to N3. The limit and all terms are integrable; all bounds hold on one full-measure set.

Receiving activity. N3: Integral convergence

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: L^1 error bound; prior exports E1/v1, E2/v1

T3: N2/E2/v1 to N3. $\|f_n - f\|_1 \to 0$.

Receiving activity. N3: Integral convergence

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: L^1 error bound; prior exports E1/v1, E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Common full-measure domain

Object and role. Remove all exceptional null sets at once.

Inputs. Measurable f_n; a.e. convergence; a.e. bounds $|f_n| \le g$ with $g \ge 0$ integrable; foundational imports stated in §1

C — Contract

Own E1/v1: The limit and all terms are integrable; all bounds hold on one full-measure set.

K — Knowledge

Retain Measurable f_n; a.e. convergence; a.e. bounds $|f_n| \le g$ with $g \ge 0$ integrable; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Remove all exceptional null sets at once.

V — Validation

A separate exceptional set for each n must be consolidated.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Take the union of the convergence exceptional set and the countably many bound exceptions.

N1.2 It is null.

N1.3 On its complement every bound holds and $f_n \to f$; redefine f and f_n on that null set if needed.

N1.4 A pointwise limit of measurable functions is measurable, and $|f| \le g$ by continuity of absolute value.

N1.5 Thus $\int |f| \le \int g < \infty$; every f_n is integrable too.

Export E1/v1. The limit and all terms are integrable; all bounds hold on one full-measure set.

Acceptance gate. A separate exceptional set for each n must be consolidated. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: A separate exceptional set for each n must be consolidated. Recheck dependent nodes N2, N3.

B. N2/v1: L^1 convergence from Fatou

Object and role. Apply Fatou to a nonnegative error complement.

Inputs. Integrable f and bounds; $h_n = 2g - |f_n - f|$; prior exports E1/v1

C — Contract

Own E2/v1: $\|f_n - f\|_1 \to 0$.

K — Knowledge

Retain Integrable f and bounds; $h_n = 2g - |f_n - f|$; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Apply Fatou to a nonnegative error complement.

V — Validation

Avoid $\infty - \infty$; finiteness of the dominating integral is used explicitly.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 The triangle inequality gives $0 \le |f_n - f| \le 2g$, so $h_n \ge 0$ and $h_n \to 2g$ almost everywhere.

N2.2 Fatou yields $2\int g \le \liminf(2\int g - \int |f_n - f|) = 2\int g - \limsup \int |f_n - f|$.

N2.3 All quantities subtracted are finite because $g \in L^1$.

N2.4 Hence the nonnegative error integrals have limsup0 and converge to 0.

Export E2/v1. $\|f_n - f\|_1 \to 0$.

Acceptance gate. Avoid $\infty - \infty$; finiteness of the dominating integral is used explicitly. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Avoid $\infty - \infty$; finiteness of the dominating integral is used explicitly. Recheck dependent nodes N3.

C. N3/v1: Integral convergence

Object and role. Control the integral difference by its absolute error.

Inputs. L^1 error bound; prior exports E1/v1, E2/v1

C — Contract

Own E3/v1: $\int f_n \to \int f$, with $f \in L^1$ and L^1 convergence.

K — Knowledge

Retain L^1 error bound; prior exports E1/v1, E2/v1; keep their assumptions and versions attached.

S — Strategy

Control the integral difference by its absolute error.

V — Validation

Pointwise convergence alone is not the gate for an integral exchange.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Linearity and the triangle inequality imply $|\int f_n - \int f| = |\int (f_n - f)| \le \int |f_n - f| \to 0$.

N3.2 This proves the full contract, including integrability of the limit.

N3.3 The proof does not claim pointwise convergence at exceptional points or uniform convergence.

Export E3/v1. $\int f_n \to \int f$, with $f \in L^1$ and L^1 convergence.

Acceptance gate. Pointwise convergence alone is not the gate for an integral exchange. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Pointwise convergence alone is not the gate for an integral exchange. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Common full-measure domain

1. Take the union of the convergence exceptional set and the countably many bound exceptions.
2. It is null.
3. On its complement every bound holds and $f_n \to f$; redefine f and f_n on that null set if needed.
4. A pointwise limit of measurable functions is measurable, and $|f| \le g$ by continuity of absolute value.
5. Thus $\int |f| \le \int g < \infty$; every f_n is integrable too.

Define $h_n = 2g - |f_n - f| \ge 0$ after modifying on the common measurable null set.

B. §4.N2 — L^1 convergence from Fatou

1. The triangle inequality gives $0 \le |f_n - f| \le 2g$, so $h_n \ge 0$ and $h_n \to 2g$ almost everywhere.
2. Fatou yields $2\int g \le \liminf(2\int g - \int |f_n - f|) = 2\int g - \limsup \int |f_n - f|$.
3. All quantities subtracted are finite because $g \in L^1$.
4. Hence the nonnegative error integrals have limsup0 and converge to 0.

C. §4.N3 — Integral convergence

1. Linearity and the triangle inequality imply $|\int f_n - \int f| = |\int (f_n - f)| \le \int |f_n - f| \to 0$.
2. This proves the full contract, including integrability of the limit.
3. The proof does not claim pointwise convergence at exceptional points or uniform convergence.

□

V. WORKED ORCHESTRATION

On[0,1], $f_n(x) = x^n$ is dominated by 1 and converges to 0 except at $x = 1$. Thus $f = 0$ as an L^1 class; $\int x^n = 1/(n+1) \to 0$, matching the theorem. Uniform convergence fails because the supremum remains1.

A. Boundary and hypothesis audit

$On(0, 1)$, $f_n = n \cdot 1_{0,1/n}$ converges pointwise to 0 while $\int f_n = 1$. If an L^1 dominator existed DCT would contradict this calculation; directly its required lower bounds force nonintegrable growth near0. Validation blocks the common-dominator claim rather than modifying the limit calculation.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

$On(0,1)$, $f_n = n \cdot 1_{0,1/n}$ converges pointwise to 0 while $\int f_n = 1$. If an L^1 dominator existed DCT would contradict this calculation; directly its required lower bounds force nonintegrable growth near0. Validation blocks the common-dominator claim rather than modifying the limit calculation.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Fatou and dominated convergence: MIT Measure and Integration.